



SQL DB with PostgreSQL

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Agenda

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3. Key Features
4. Example Usage
5. PostgreSQL vs MySQL



Intro

- What is a SQL Database?
 - A databases that organizes data into tables composed of rows and columns, forming relations between these tables using a shared value (e.g., an ID)
- What is PostgreSQL?
 - PostgreSQL is an object-relational database that supports relational SQL querying and non-relational JSON querying



History and Background

- Began as a project at UC Berkeley in 1986 as an evolution of the Ingres database project; originally names Postgres
- Designed with the goal of adding as few features as needed to support multiple data types
- Renamed to PostgreSQL in 1996 to make clear its support for SQL
- Now maintained as an open source project by a global community known as the PostgreSQL Global Development Group



Key features and Benefits

- Supports advanced data types, such as Enums and Geometry
- ACID compliant: the database ensures atomicity, consistency, isolation, and durability, making database transactions reliable
- PostGIS: An extension of the database that is optimized for Geospatial coordinates
- Supports both relational and non-relational queries
- Extendable and Flexible: Easy to add custom functions, operators and data types due to its object oriented features



Examples of its usage and Relation to Capstone

- A geospatial database that stores object data projected onto a map
- An Online Transaction Processing Database where the accurate modification of monetary data is crucial
- A db for a social media platform that stores profiles, posts, and more.
- Example code:
 - `CREATE TABLE cities (`
 - `name text,`
 - `population real,`
 - `elevation int -- (in ft)`
 - `);`
 - `CREATE TABLE capitals (`
 - `state char(2) UNIQUE NOT NULL`
 - `) INHERITS (cities);`



PostgreSQL vs MySQL

- MySQL is an alternative relational database option
- It is often perceived as more lightweight and faster to learn than PostgreSQL, while PostgreSQL is more flexible and can “scale faster to a very large volume”



Sources

<https://www.percona.com/blog/what-is-postgresql-used-for/>

<https://aws.amazon.com/rds/postgresql/what-is-postgresql/>

<https://www.postgresql.org/docs/current/tutorial-inheritance.html>